

# LAB 4 — TREASURE HUNT

## AI-Assisted Development Chat Record

This PDF records the relevant AI-assisted development conversation for the Treasure Hunt Pygame Lab 4 work. It preserves the main requests, debugging steps, implementation decisions, and submission preparation discussed in the chat.

# 1. Project Inspection and Task 1

**USER**

I am working on the existing Treasure Hunt Pygame project. Do not rewrite the whole project or change unrelated functionality. First inspect the existing code structure and explain briefly how the dungeon grid, player starting position, and collision/update logic currently work. Then implement Task 1 from the README: add floor traps that send the player visibly back to the starting position when stepped on, and display a clear status message such as "Trap! Back to start!".

**AI ASSISTANT**

The README confirmed Task 1: add floor traps that send the player back to their starting position and show a clear status message. The assistant explained that the actual game source was needed before making exact code edits.

**USER**

I uploaded game.py. Where is the KEY/CHEST drawing code?

**AI ASSISTANT**

The assistant identified the drawing section inside draw(), around the KEY and CHEST rendering, and explained that the TRAP drawing block should be added immediately after the CHEST block.

**USER**

Okay what do I do after this?

**AI ASSISTANT**

The assistant instructed adding TRAP to the tile constants and adding TRAP\_COUNT, then generating traps on walkable floor cells outside the starting room.

**USER**

I'm done.

**AI ASSISTANT**

The assistant instructed saving and running the game, then checking for visible traps, trap reset behavior, the trap status message, key/chest behavior, and restart behavior.

## 2. Debugging the Existing Code

**USER**

I tried python game.py and got: can't open file ... No such file or directory.

**AI ASSISTANT**

The assistant explained that the terminal was in C:\Users\kaila instead of the project directory. It identified the project folder from the VS Code screenshot as C:\Users\kaila\OneDrive\Desktop\24\_treasure\_hunt-main and instructed changing into that folder before running python game.py.

**USER**

I got an IndentationError around if len(rooms) >= 2.

**AI ASSISTANT**

The assistant explained that the error was caused by inconsistent indentation and provided a clean replacement for the generate\_world() section, using consistent spaces.

**USER**

I got another indentation error around if cell == KEY.

**AI ASSISTANT**

The assistant identified the draw() section as having duplicated/mis-indented trap and chest code and instructed replacing the affected draw() section with one clean block.

**USER**

Here is my entire code.

**AI ASSISTANT**

After reviewing the supplied code, the assistant identified the duplicated draw() code, incorrect indentation, missing start\_pos storage, and missing trap collision handling. It then offered a complete corrected game.py.

### 3. Moving from Task 1 to Tasks 2–4

**USER**

This is what I am supposed to do: A PDF of code changes for all the tasks, the AI chat saved as a PDF, a video of the game before fixing the tasks, a video of the game after fixing all the tasks, and the final updated code, all pushed to GitHub in lab4 folder.

**AI ASSISTANT**

The assistant converted this into a submission checklist: a code-changes PDF, AI-chat PDF, before video, after video, final code, and a lab4 folder in GitHub. It also explained that Tasks 1–4 had to be completed before final evidence was prepared.

**USER**

Okay do task 2 task 3 and 4.

**AI ASSISTANT**

The assistant used the Lab 4 README requirements: Task 2 is an enemy guard that patrols back and forth near the chest and resets the player on collision; Task 3 is a real-time mini-map showing walls, floor, and player position; Task 4 is an inventory slot that is empty initially and shows a key icon after pickup.

**AI ASSISTANT**

A complete updated game.py was then provided. It added a Guard class with two patrol points, guard collision/reset behavior, a real-time mini-map, and an inventory slot with a drawn key icon while retaining the existing procedural dungeon, movement, key/chest, and restart functionality.

## 4. Testing and Submission Preparation

**USER**

I'm done with the tasks.

**AI ASSISTANT**

The assistant explained that the next stage was submission preparation: record the before and after videos, create the code-changes PDF, save the AI chat as a PDF, and place everything in the GitHub lab4 folder.

**USER**

Okay let's do the first one.

**AI ASSISTANT**

The assistant generated the first submission document, 1\_Code\_Changes\_Lab4.pdf. It documents Tasks 1–4, the major code changes, and the testing checklist.

**USER**

Okay next.

**AI ASSISTANT**

This PDF is the second submission document: the AI-assisted development chat record for Lab 4.

End of AI chat record. The full ChatGPT conversation remains the primary source of the AI-assistance history; this PDF is a submission-ready record of the relevant Lab 4 exchange.